

PHY102 Electricity and Magnetism Jan-April 2026: Assignment 8

Arvind, IISER Mohali, arvind@iisermohali.ac.in

To be discussed in the Tutorial on April 10, 2026

1. Consider an infinite wire passing through the origin and along the z -axis. A current I flows through this wire along the direction $\hat{z}$. (a) Calculate the vector potential $\vec{A}$ for this current and then by taking $\nabla \times \vec{A}$ calculate the magnetic field $\vec{B}$. (b) Using the Biot-Savart law, calculate the magnetic field directly.
2. Consider a square loop of size a placed in the $x - z$ plane with one corner at the origin in the magnetic field of the wire described in Problem 1. The loop is moving with a uniform velocity $\vec{v} = v\hat{x}$ along the x -axis. Calculate the rate of change of flux of the magnetic field through this loop and also calculate the induced emf.
3. Consider a circular loop of radius r in the $x - y$ plane with its center at the origin and carrying a current I in the clockwise direction. Calculate the magnetic field along the z axis by using the Biot-Savart law.
4. Consider a vector field $\vec{B} = kx\hat{x} + ky\hat{y} - 2kz\hat{z}$ (a) Show that this field is divergence-free. If this was to be a valid magnetic field what will be corresponding current density? Now imagine a hemisphere of radius R . The hemisphere has a flat part which is a disc of radius R . Show that the flux of the field $\vec{B}$ through the flat disc surface is equal to the flux through the hemispherical surface.